

SUPPLY CHAIN DELIVERY PERFORMANCE MONITORING SYSTEM

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Project Focus: Supply chain data processing and delivery performance monitoring

Case Study Organisation: Zenvorq Freightways Solutions (Pty) Ltd
Fictional logistics company used for this project scenario

INTRODUCTION

Software Engineering team at Zenvorq received reports from the warehouse operations team indicating that parcel-processing times have become inconsistent. Some parcels move through the sorting facility within minutes, while others remain in the processing queue for significantly longer.

The company currently relies on historical order records to evaluate delivery performance. **However, management needs a more structured way to identify delivery delays, compare shipping methods, understand performance across different markets and determine where operational improvements may be needed**

Task: *Your team lead has assigned you to develop a prototype Supply Chain Delivery Performance Monitoring System using an existing historical supply-chain dataset*.

Objective: to transform raw supply-chain data into useful operational insights that help management identify delivery-performance problems and make evidence-based decisions.

Tools	IntelliJ IDE
Java	CSV File
Python	Java Collection Framework
Numpy	Matplotlib
Pandas	

Dataset: DataCoSupplyChainDataset
https://www.kaggle.com/datasets/saicharankomati/dataco-supply-chain-dataset

The dataset contained 180,519 records and 53 columns. Seven relevant fields were selected: Order ID, actual shipping days, scheduled shipping days, delivery status, shipping mode, market and order region.

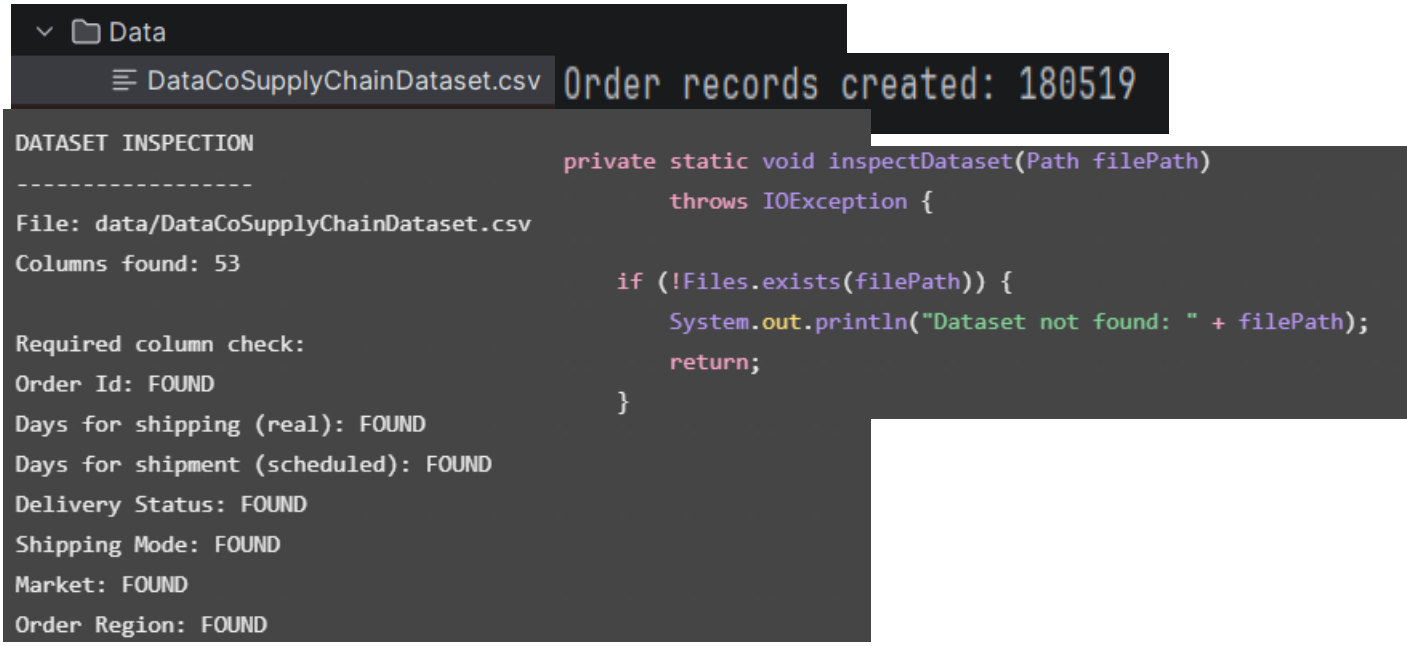
The dataset used Windows-1252 encoding, which was configured in Java to ensure that the source file could be read successfully.

TASK 1: Build a supply-chain data processing application

Data Preparation and Processing

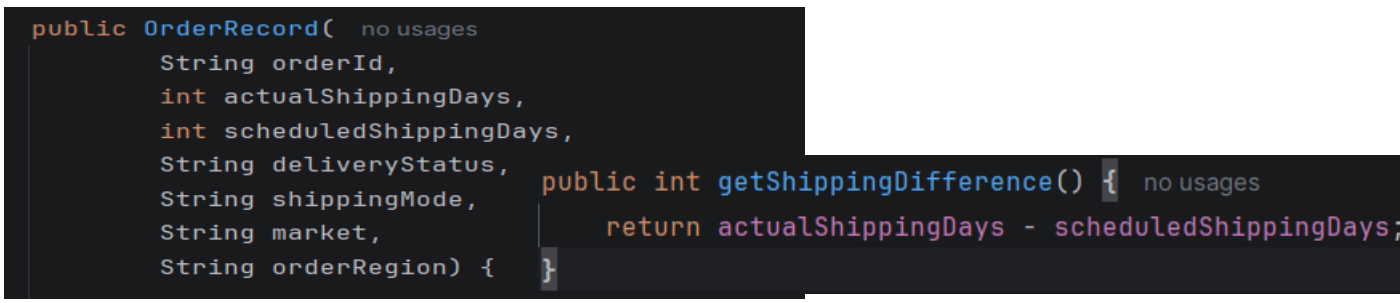
The supplied CSV file was placed in the project's data directory. A Java inspection routine was developed to read the file, examine its column headers, confirm that the seven required fields were available, display sample records and count the dataset rows.

The dataset contained 180,519 records, and all required columns were found. Windows-1252 encoding was used because the original file could not be read successfully using UTF-8**.**



Calculating Shipping Difference

An OrderRecord class was created to represent an individual order. It stores the order identifier, actual and scheduled shipping durations, delivery status, shipping mode, market, order region and calculates the difference between actual and scheduled shipping days.



Reading and Processing Records

The first loaded record had an order ID of 77202, an actual shipping duration of three days and a scheduled duration of four days. The calculated difference was **−1 day**, indicating that the actual duration was one day shorter than scheduled.

```
private static List<OrderRecord> loadOrders(Path filePath) throws IOException {

    List<OrderRecord> orders = new ArrayList<>();
    int validRecords = 0;
    int invalidRecords = 0;
```

```
Order ID: 77202
Actual shipping days: 3
Scheduled shipping days: 4
Shipping difference: -1
```

Queue Implementation

A Java Queue implemented using LinkedList was introduced to demonstrate first-in, first-out (FIFO) order processing. The records were added to the queue, and the poll() method retrieved and removed the first order.

```
Queue<OrderRecord> orderQueue = new LinkedList<>(orders);
OrderRecord nextOrder = orderQueue.poll();
```

```
First order processed from queue:
Order ID: 77202
Shipping difference: -1
Orders remaining in queue: 180518
```

Data Validation

Validation rules were added to check for empty order identifiers, negative shipping durations and missing required text fields. Records that failed these checks were counted as invalid, while valid records were retained for further processing.

```
if (orderId.isBlank()
        || actualDays < 0
        || scheduledDays < 0
        || deliveryStatus.isBlank()
        || shippingMode.isBlank()
        || market.isBlank()
        || orderRegion.isBlank()) {

    invalidRecords++;
    System.out.println(
            "Invalid data at line " + lineNumber);
    continue;
}
```

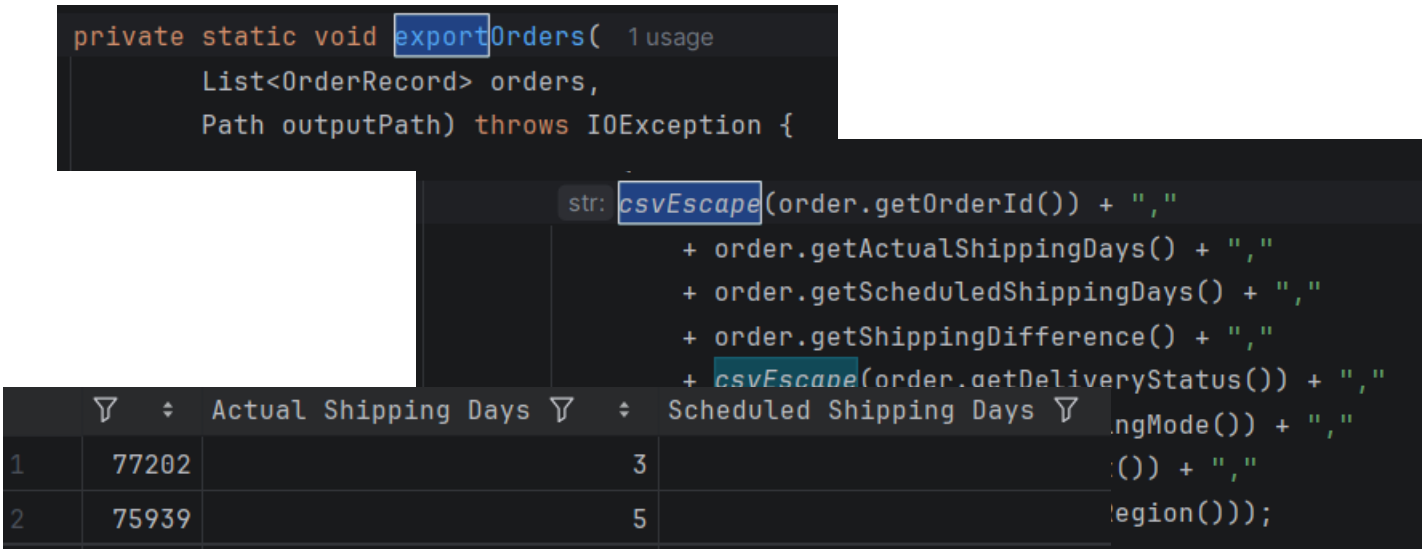
```
for (String column : requiredColumns) {
    if (!columnIndexes.containsKey(column)) {
        throw new IOException(
                "Required column is missing: " + column);
    }
}
```

```
Validation Summary
------------------
Valid records: 180519
Invalid records: 0
Total processed: 180519
Exported records: 180519
```

Exporting the Processed Dataset

An export method was implemented to write the processed records to output/processed_orders.csv. The exported file contains eight columns: the seven selected fields and the calculated shipping difference.

The program successfully exported 180,519 records. The output file uses UTF-8 encoding to support subsequent analysis.



Conclusion

Task 1 established a Java-based process for inspecting, loading, validating and exporting supply chain delivery records. The application successfully processed 180,519 valid records and generated a reduced CSV containing the selected order fields and calculated shipping differences. The resulting dataset is prepared for the next stage of analysis and visualisation using Python.

TASK 2 : Analyse Delivery Perforance and Analyze Delays

Importing relevant libraries and loading the dataset

Loaded output/processed_orders.csv into a Pandas DataFrame using Python in Jupyter Notebook

```
file_path = "processed_orders.csv"

df = pd.read_csv(file_path)

print("Dataset loaded successfully.")
print("Number of records:", len(df))
print("Number of columns:", len(df.columns))
```

```
Dataset loaded successfully.
Number of records: 180519
Number of columns: 9
```

Calculate shipping statistics

Calculated the average and median actual shipping duration and compared actual shipping days with scheduled shipping days.

```
average_shipping_days = df["Actual Shipping Days"].mean()
median_shipping_days = df["Actual Shipping Days"].median()

print(f"Average actual shipping duration: {
print(f"Median actual shipping duration: {
```

```
Average actual shipping duration: 3.50 days
Median actual shipping duration: 3.00 days
```

```
Average actual shipping duration: 3.50 days
Average scheduled shipping duration: 2.93 days
Average difference: 0.57 days
Orders shipped later than scheduled: 103,400
Orders shipped on schedule: 33,753
Orders shipped earlier than scheduled: 43,366
```

Analysing Delivery Performance

Compared actual and scheduled shipping days to classify deliveries as early, late, or on schedule. Also calculated the percentage of orders recorded as late deliveries.

```
delivery_status = df["Delivery Status"].astype(str).str.strip().str.lower()
late_delivery_count = (delivery_status == "late delivery").sum()
total_records = len(df)
late_delivery_percentage = (late_delivery_count / total_records) * 100
```

```
Total records: 180,519
Records marked as late deliveries: 98,977
Percentage of late deliveries: 54.83%
```

Compared delivery performance across shipping modes, markets and regions

Grouped the data by shipping mode to compare delivery performance. First Class had the highest recorded late-delivery rate at 95.32%, followed by Second Class at 76.63%. Compared late-delivery rates across markets and regions. Europe had the highest market-level rate at **55.21%**, while Central Africa had the highest regional rate at **57.96%**.

DELIVERY PERFORMANCE BY SHIPPING MODE

Shipping Mode	Total_Orders	Average_Actual_Days	Average_Scheduled_Days	Average_Shipping_Difference	Late_Deliveries	Late_Delivery_Percentage
First Class	27814	2.00	1.0	1.00	26513	95.32
Second Class	35216	3.99	2.0	1.99	26987	76.63

DELIVERY PERFORMANCE BY MARKET

Market	Total_Orders	Average_Actual_Days	Average_Scheduled_Days	Average_Shipping_Difference	Late_Deliveries	Late_Delivery_Percentage
Europe	50252	3.49	2.92	0.57	27743	55.21

DELIVERY PERFORMANCE BY ORDER REGION

Order Region	Total_Orders	Average_Actual_Days	Average_Scheduled_Days	Average_Shipping_Difference	Late_Deliveries	Late_Delivery_Percentage
Central Africa	1677	3.56	2.92	0.64	972	57.96

Analysing Shipping Delays and Order Profit

Examined the relationship between shipping difference and order profit. The correlation was approximately -0.0054, indicating almost no linear relationship between these two variables

```
df["Shipping Difference"] = pd.to_numeric(
    df["Shipping Difference"], errors="coerce"
)

df["Order Profit Per Order"] = pd.to_numeric(
    df["Order Profit Per Order"], errors="coerce"
)
```

```
Correlation between shipping difference and order profit: -0.0054
```

Checking Data Quality

Checked the dataset for missing values, exact duplicate rows, shipping-difference calculation errors, and delivery-status inconsistencies.

```
MISSING VALUES BY COLUMN
------------------------
Order ID                  0
Actual Shipping Days      0
Scheduled Shipping Days   0
Shipping Difference       0
Order Profit Per Order    0
Delivery Status           0
Shipping Mode             0
Market                    0
Order Region              0
Delivery Timing           0
dtype: int64

SHIPPING DIFFERENCE VALIDATION
------------------------------
Records with incorrect shipping differences: 0
```

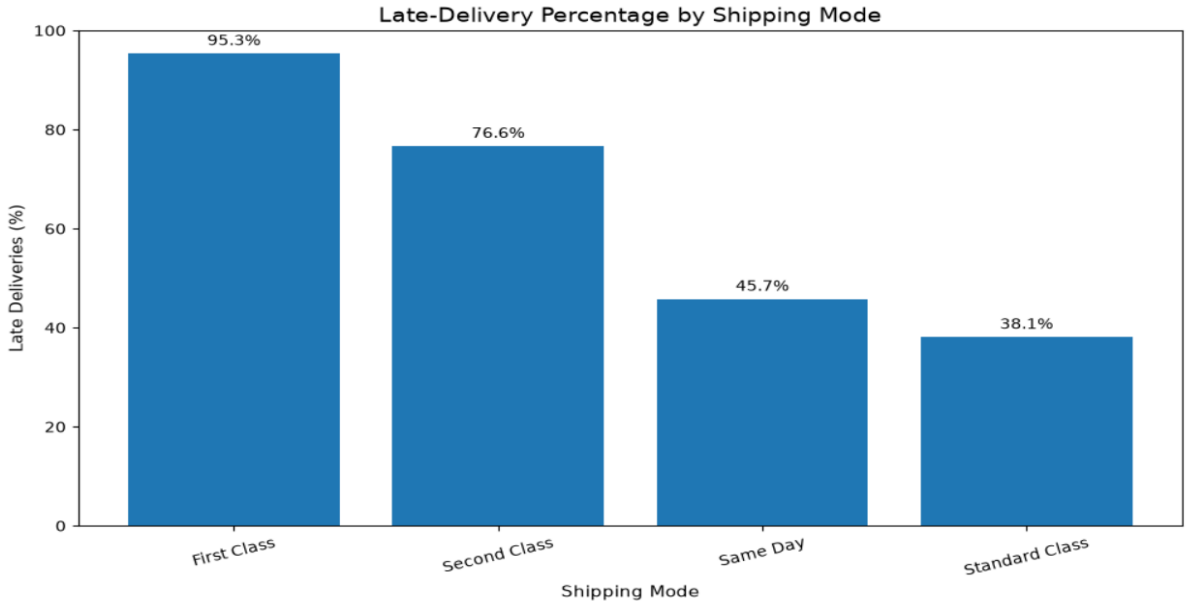
```
DATA QUALITY CHECKS
-------------------
Total records: 180,519
Duplicate rows: 60
Repeated Order IDs: 114,767
```

The 60 duplicate rows were identified for further investigation rather than automatically removed.

TASK 3 : Develop delivery-performance visualisations

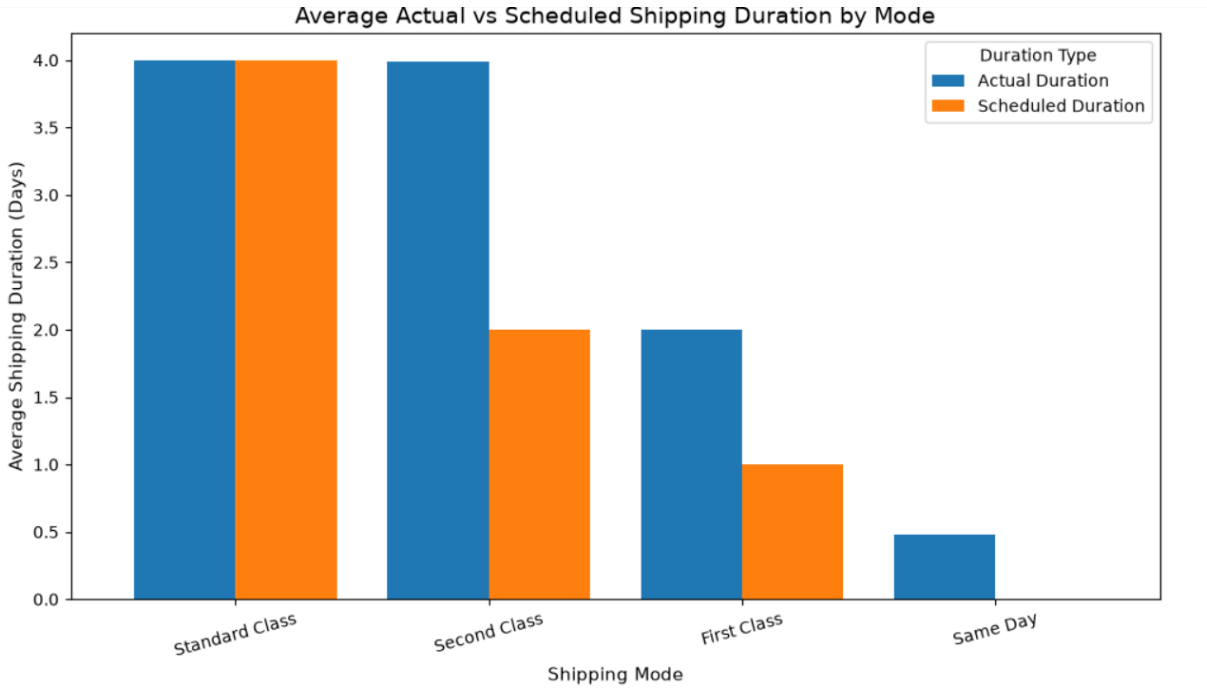
Late-Delivery Rate by Shipping Mode

Created a bar chart comparing late-delivery rates across shipping modes. First Class recorded the highest rate at **95.32%**.



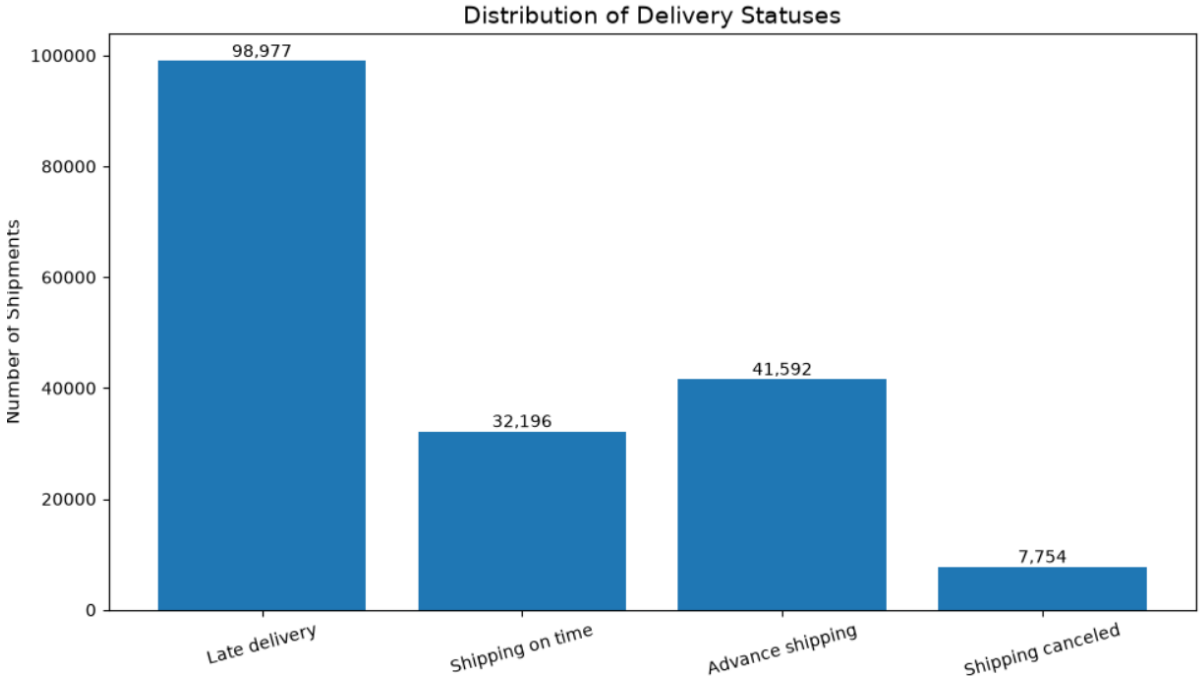
Actual vs Scheduled Shipping Duration

Created a grouped bar chart comparing average actual and scheduled shipping durations for each shipping mode.



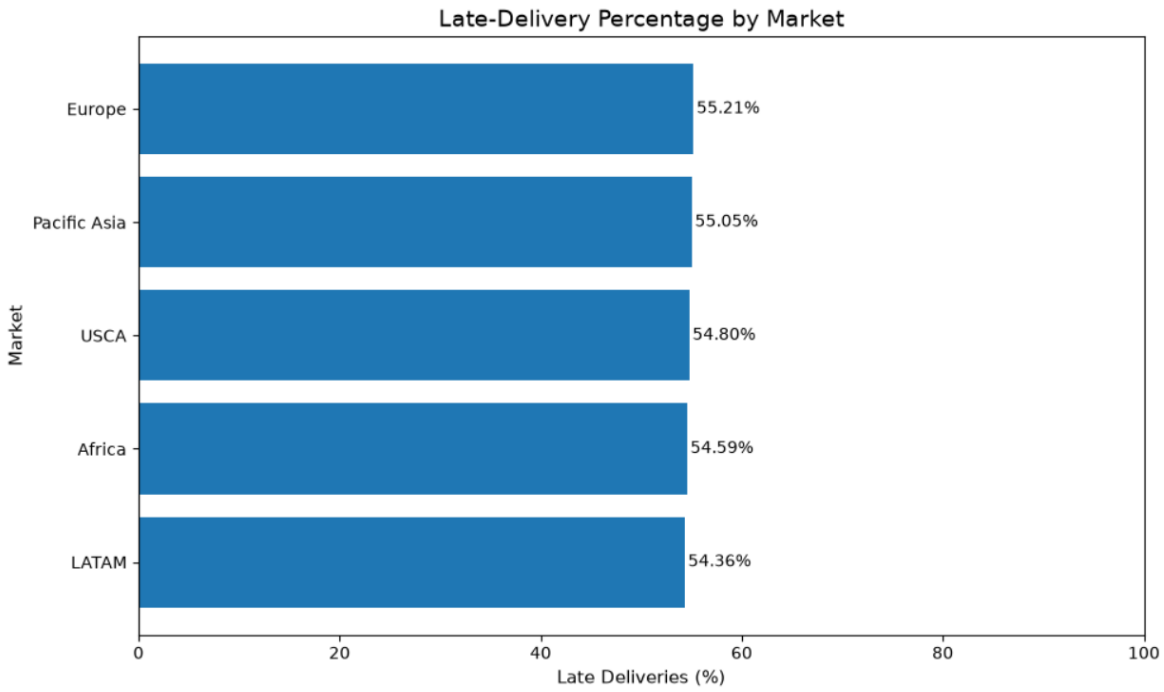
Delivery Status Distribution

Created a bar chart showing the number of records in each delivery status: Late delivery, Shipping on time, Advance shipping, and Shipping canceled.



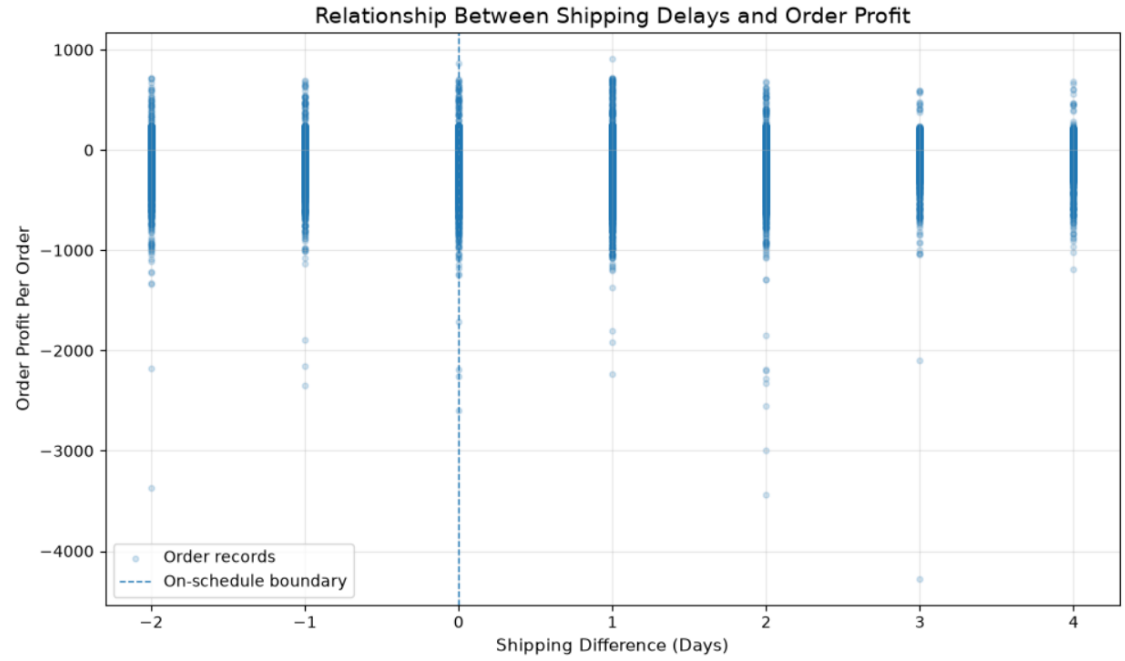
Late-Delivery Rate by Market

Created a horizontal bar chart comparing late-delivery rates across markets. Europe recorded the highest rate at **55.21%**.



Shipping Difference vs Order Profit

Created a scatter plot to examine the relationship between shipping difference and order profit. The correlation was approximately **-0.0054**, indicating almost no linear relationship.



Five visualisations were generated and saved as PNG files in the output/charts/ directory to support the analysis and communicate delivery-performance patterns

TASK 4: BUSINESS QUESTIONS, FINDINGS AND RECOMMENDATIONS

What proportion of orders is marked as late?

Of the 180,519 records, 98,977 were marked as late deliveries, representing 54.83% of the dataset.

2. Which shipping modes have the highest late-delivery rates?

First Class had the highest late-delivery rate at 95.32%, followed by Second Class at 76.63%, Same Day at 45.74%, and Standard Class at 38.07%.

3. How much longer do shipments take compared with their scheduled durations?

Average actual shipping duration was 3.50 days, compared with an average scheduled duration of 2.93 days. Shipments therefore took an average of 0.57 days longer than scheduled**.**

4. Which markets or regions show weaker delivery performance?

Europe had the highest late-delivery rate among markets at 55.21%, followed by Pacific Asia at 55.05%. Central Africa recorded the highest regional rate at 57.96%.

5. Do orders with longer shipping durations also tend to have lower profits?

The correlation between shipping difference and order profit was approximately -0.0054, indicating almost no linear relationship. The analysis does not establish that longer shipping durations cause lower profits.

6. What additional information would management need to determine the causes of delays?

Management would benefit from carrier and route identifiers, dispatch and carrier pickup timestamps, warehouse processing times, expected and actual delivery dates, delivery-exception and cancellation reasons, and relevant shipping cost and

order-value information. These details would help identify where and why delays occur.

CONCLUSION

The analysis found that 54.83% of records were marked as late deliveries, with First Class and Second Class showing the highest late-delivery rates. Actual shipping duration exceeded scheduled duration by an average of 0.57 days. Europe and Central Africa showed weaker delivery performance, while shipping difference had almost no linear relationship with order profit. Further operational data is required to establish the causes of delays.
